

Assignment 4 – Fingerprint Quality Assessment & Scoring Pipeline Report

Name: _____

Objective: Implement a multi-metric quality assessment pipeline for contactless fingerprint images using OpenCV and Streamlit.

Implemented Metrics: Blur detection (Laplacian variance), Brightness check (mean grayscale), Glare detection (overexposed pixel ratio), ROI completeness (finger area), Ridge clarity (Gabor filter response).

Composite Quality Score: The five metrics are combined into a quality score from 0–100. Images scoring 60 or above are accepted for processing.

User Interface: A Streamlit application displays the uploaded image, quality metrics, processing time, overall decision (PASS/ACCEPTABLE/FAIL), reasons, and suggestions.

Testing: The application was tested using 20 images (5 good, 5 blurry, 5 dark, and 5 glare). The quality gate correctly identifies common capture issues and provides guidance for retaking images.

Report Answers

1. Blur threshold: Calibrated experimentally using the collected dataset and Laplacian variance.
2. Ridge clarity was the most challenging because selecting a stable threshold required testing on multiple images.
3. NFIQ2 is a fingerprint quality standard designed for contact scanners; it is less reliable for contactless mobile captures because of lighting, perspective, and resolution differences.
4. Additional checks: wet finger, excessive finger angle, finger too far from the camera.
5. For naturally worn fingerprints, the system should adapt thresholds or request multiple captures instead of rejecting users immediately.

Conclusion: The developed quality assessment pipeline successfully evaluates fingerprint image quality and provides actionable feedback before biometric processing.